

Chapter 1

INTRODUCTION

- 1.1* Purpose and Scope
- 1.2* Problem Statement
- 1.3* Objectives

1.1 Introduction

In today's fast-paced and technology-driven world, the healthcare sector is under increasing pressure to deliver efficient, accurate, and timely services. Traditional hospital management methods, which often rely on manual paperwork and fragmented data systems, are no longer sufficient to meet the growing demands of patients and healthcare providers. To address these challenges, the **Hospital Management System (HMS)** has been developed as an integrated software solution designed to digitize and streamline hospital operations.

This project focuses on designing and implementing a user-friendly system that simplifies core administrative and medical processes such as patient registration, appointment scheduling, staff management, billing, inventory control, and electronic medical records. By centralizing data and automating repetitive tasks, the system reduces human error, saves time, and enhances the overall efficiency of hospital workflows.

1.2 Purpose & Scope

The purpose of running the hospital management system are listed as :

- a. **To automate hospital operations** such as patient registration, appointment scheduling, billing, and record management.
- b. **To reduce manual errors** and improve the accuracy of patient and hospital data.
- c. **To enhance communication** between different departments (reception, doctors, pharmacy, labs, etc.).
- d. **To maintain electronic medical records (EMR)** for quick retrieval and better diagnosis history tracking.
- e. **To improve patient experience** by reducing waiting time and streamlining processes.

Similarly, the scope of an HMS is to automate and streamline administrative and clinical processes, improve patient care coordination, enhance data security and accuracy. The detailed scope of HMS are listed :

- a. **User Roles:** Supports multiple user types such as Administrator, Doctor, Nurse, Receptionist, and Patient.
- b. **Patient Module:** Registration, admission, discharge, appointment scheduling, and access to medical history.

- c. **Doctor Module:** View schedules, patient information, prescriptions, and lab reports.
- d. **Billing Module:** Automated invoice generation, payment tracking, and financial reporting.
- e. **Pharmacy Management:** Inventory control, medicine issue records, and stock alerts.

1.3 Problem Statement

Here are several well-articulated **Problem Statements** for your **Hospital Management System (HMS)** project. These identify key issues that the system is designed to solve:

a) **Inefficient Manual Processes:**

Many hospitals still rely on manual methods for managing patient records, scheduling appointments, and handling billing. This leads to delays, human errors, and a lack of real-time data access.

b) **Fragmented Data Management:**

Information is often scattered across different departments without a unified system, making it difficult to track a patient's complete history or generate accurate reports.

c) **Limited Communication Between Departments:**

A lack of integration between departments (e.g., pharmacy, labs, wards, and billing) causes delays in service delivery and increases the chances of miscommunication.

d) **Time-Consuming Patient Handling:**

Long queues for appointments, billing, and report collection negatively impact patient satisfaction and increase the workload for hospital staff.

e) **Poor Record-Keeping and Data Retrieval:**

Paper-based records are difficult to maintain, store, and retrieve. They are also vulnerable to loss, damage, or unauthorized access.

1.4 Objectives

A **hospital management system** is an integrated and robust platform designed to connect and manage hospitals across all departments, such as medical, financial, patient, and administration, in a single place.

The key objectives performed under system of hospital management are as :

- Computerize all the details of the patient and hospital.
- Appointment scheduling of patients with doctors for mutual convenience.
- To perform automation of workflows.
- Secure hospital data and enhance retrievability as it contains sensitive information related to patients, staff, and hospital.
- Design a system to improve patient experience.

So, the HMS takes care of all the requirements of the hospitals and can provide **easy and effective storage of information related to patients who come to the hospital for better patient care service.** It is a complete hospital suite serving all functional areas of the hospital. It covers the cycle of hospital workflow from appointment, patient registration, patient history, patient care, and doctor.

Chapter 2

SYSTEM ANALYSIS

2.1.1 Requirement Analysis

- i.* Functional Requirements
- ii.* Non Functional Requirements

2.1.2 Feasibility Analysis

2.1.3 System Modeling

2.1 System Analysis

System analysis is a crucial phase in the development of the Hospital Management System, involving the in-depth study and understanding of the existing hospital processes and identifying the requirements for building an efficient, automated system. The goal is to understand how the current system works, what problems exist, and how a new system can be designed to address those issues effectively. It serves as the foundation for designing a system that improves hospital workflow, enhances service quality, and ensures the accuracy and security of medical information.

2.1.1 Requirement Analysis

Requirement analysis is the process of identifying and documenting the functional and non-functional needs of the users to ensure that the system meets its intended purpose. For the Hospital Management System, this step is essential to ensure that all departments and users are supported by the system efficiently.

It is done under major two types :

a) Functional Requirements :

There are a lot of software requirements specifications included in the functional requirements of the Hospital Management System, which contains various processes, namely Registration, Check, Report Generation, and Database. The analysis identifies the key functions the system must perform, such as:

- Registering patients.
- Scheduling appointments.
- Managing patient records.
- Handling billing and payments.
- Generating reports.

b) Non Functional Requirements :

There are a lot of software requirements specifications included in the non-functional requirements of the Hospital Management System, which contains various processes, namely Security, Performance, Maintainability, and Reliability.

These define the system's operational attributes.

- Security and privacy of data.
- User-friendliness of the interface.
- System reliability and availability.
- Performance and response time.

2.1.2 Feasibility Analysis

A **feasibility study** is conducted to determine whether the proposed Hospital Management System is practical, achievable, and beneficial for the intended users. It evaluates various aspects of the system before the development phase begins to ensure the project is worth investing time and resources in.

A feasibility study to manage all economical, technical and legal works whether the given management completed in desired time or not can be carried out as:

- a) Technical Feasibility :** This assesses whether the current technology is capable of supporting the development and deployment of the HMS.
- i) The system will be developed using reliable technologies such as [insert technology stack, e.g., PHP & MySQL / Python & Django / Java & Oracle].
 - ii) Most hospitals already have the required hardware (PCs, network infrastructure).
 - iii) Developers and IT staff have the necessary technical skills to build and maintain the system.

Conclusion : Technically Feasible.

- b) Operational Feasibility :** This evaluates whether the system will function properly in the intended environment and whether end users will accept and use it.
- i) The system simplifies and automates daily hospital tasks, improving workflow.
 - ii) Users (doctors, nurses, admins) will be trained to use the system.
 - iii) It enhances service delivery and reduces patient wait times.
 - iv) It ensures better interdepartmental communication and data accuracy.

Conclusion: Operationally Feasible with high potential for user adoption.

- c) **Economic Feasibility** : This assesses whether the benefits of the system outweigh the costs.
- i) Initial development costs (software, hardware, training) are moderate.
 - ii) Long-term cost savings from reduced paperwork, improved efficiency, and fewer errors.
 - iii) Increased revenue from better patient management and billing accuracy.

Conclusion: Economically feasible and cost-effective in the long run.

- d) **Schedule Feasibility** : This determines whether the system can be developed within a reasonable timeframe.
- i) The development is broken into clear phases: planning, analysis, design, development, testing, deployment.
 - ii) A timeline of [insert time estimate, e.g., 3-6 months] is realistic for project completion.
 - iii) Resources and team members are available to work within the given schedule.

Conclusion: Feasible within the proposed schedule.

2.1.3 System Modeling

System Modeling is the process of visually representing the components, structure, behavior, and data flow of a system. In the Hospital Management System (HMS), system models help in understanding the overall design, user interaction, data processing, and inter-module communication. Modeling can be classified into two major approaches :

- 1) **Structural Approach** : It describe the static structure of the system — how data and components are organized.
- **Entity-Relationship Diagram (ERD)**: Shows data entities, attributes, and relationships.
 - **Data Flow Diagrams (DFDs)**: Shows the flow of data between processes and data stores.

Here's a comprehensive guide to **Data Modeling using Entity-Relationship Diagrams (ERDs)** for our **Hospital Management System (HMS)** project.

Here's a basic data model showing the key entities and their relationships:

Entities and Attributes:

1. Patient

- Attributes: P-ID, Name, DOB, Age, Gender, Mob-No
- Each patient is identified by a unique P-ID.
- A patient can have medical tests, billing information, and room assignments.

2. Doctor

- Attributes: Dept, Qualification
- A doctor is also an employee (inherited from Employee) and can consult multiple patients.

3. Nurse

- Inherits attributes from Employee
- Nurses are assigned to govern multiple rooms.

4. Receptionist

- Also a type of Employee
- Maintains Records of appointments and patient interactions.

5. Employee

- Attributes: E-ID, Name, Salary, Sex, Mob-No, Address (includes City, State, Pin-no)
- A generalized entity from which Doctor, Nurse, and Receptionist are derived.

6. Rooms

- Attributes: R-ID, Type, Capacity, Availability
- Rooms are assigned to patients and governed by nurses.

7. Bills

- Attributes: B-ID, Amount
- Associated with each patient, tracking billing transactions.

8. Test Report

- Attributes: R-ID (Report ID), Test-Type, Result
- Linked to patients, representing medical test results.

9. Records

- Attributes: Record-no, App-no
- Maintained by receptionists, these keep track of appointments and patient data.

Relationships:

1. Consults (Patient–Doctor)

Many-to-Many: A patient can consult multiple doctors, and a doctor can consult many patients.

2. Pays (Patient–Bills)

One-to-Many: A patient may pay multiple bills.

3. Has (Patient–Test Report)

One-to-Many: A patient may have multiple test reports.

4. Assigned (Patient–Rooms)

- Many-to-One: Many patients can be assigned to a room.

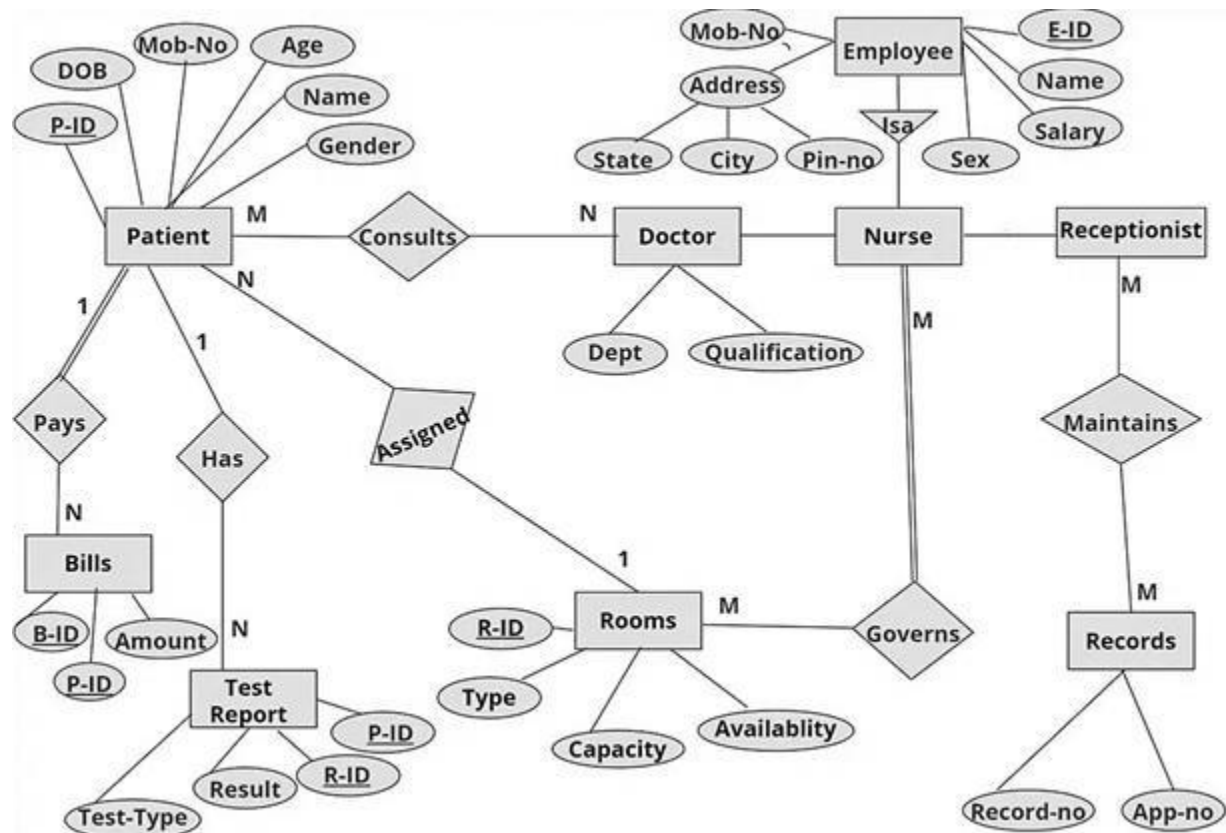
5. Governs (Nurse–Rooms)

- Many-to-Many: Nurses can govern multiple rooms, and a room may be governed by multiple nurses.

6. Maintains (Receptionist–Records)

- Many-to-Many: Receptionists maintain multiple records, and each record can be handled by multiple receptionists.

Respective ER Diagram Of Hospital Management System



DFD : A **Data Flow Diagram (DFD)** shows how data flows within a system, including

- i) External entities (sources/sinks of data)
- ii) Processes (functions or actions)
- iii) Data stores (files or databases)
- iv) Data flows (movement of data)

The respective **Context Level Diagram**(Level 0) and **DFD** Upto Level 1 for Hospital Management System are drawn as :

Context Level Diagram(Level-0) :

This is the highest level of the DFD. It represents the entire system as a single process with its interactions with external entities.

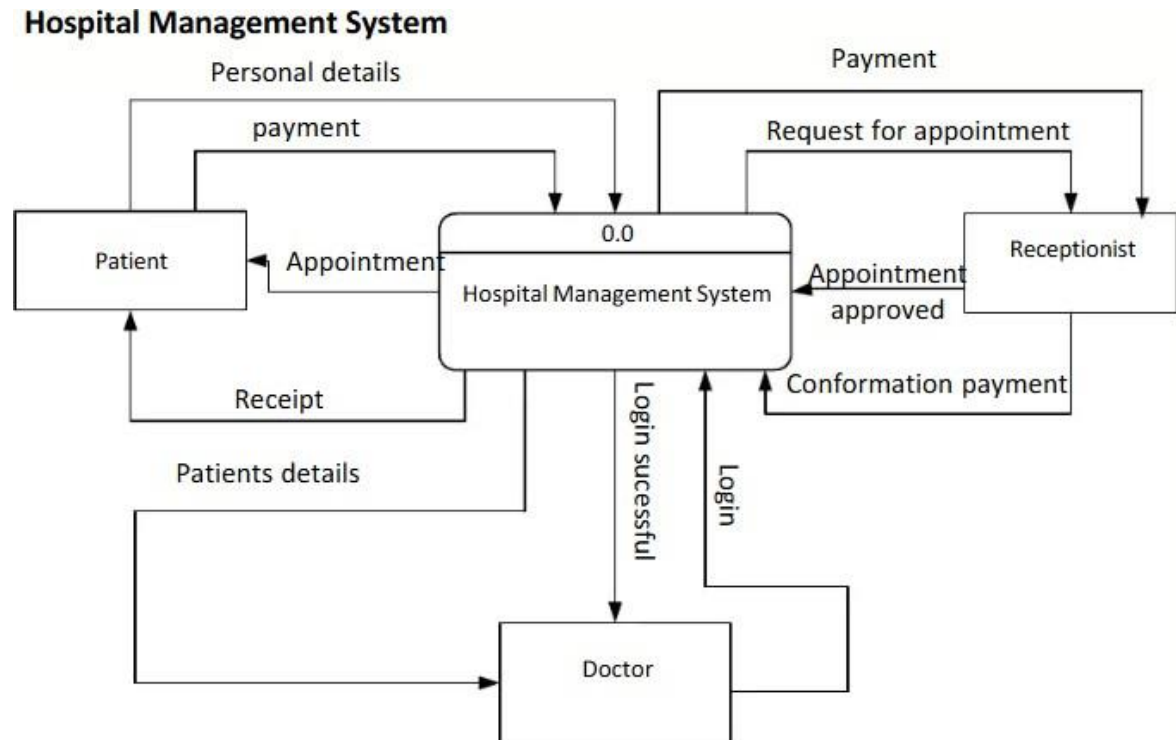


Fig : Context Level Diagram Of HMS

Level 1 DFD :

Here, Processes in level 0 (Context Diagram) can be further exploded to represents details of the sub processes. This level breaks down the main system process into sub-processes.

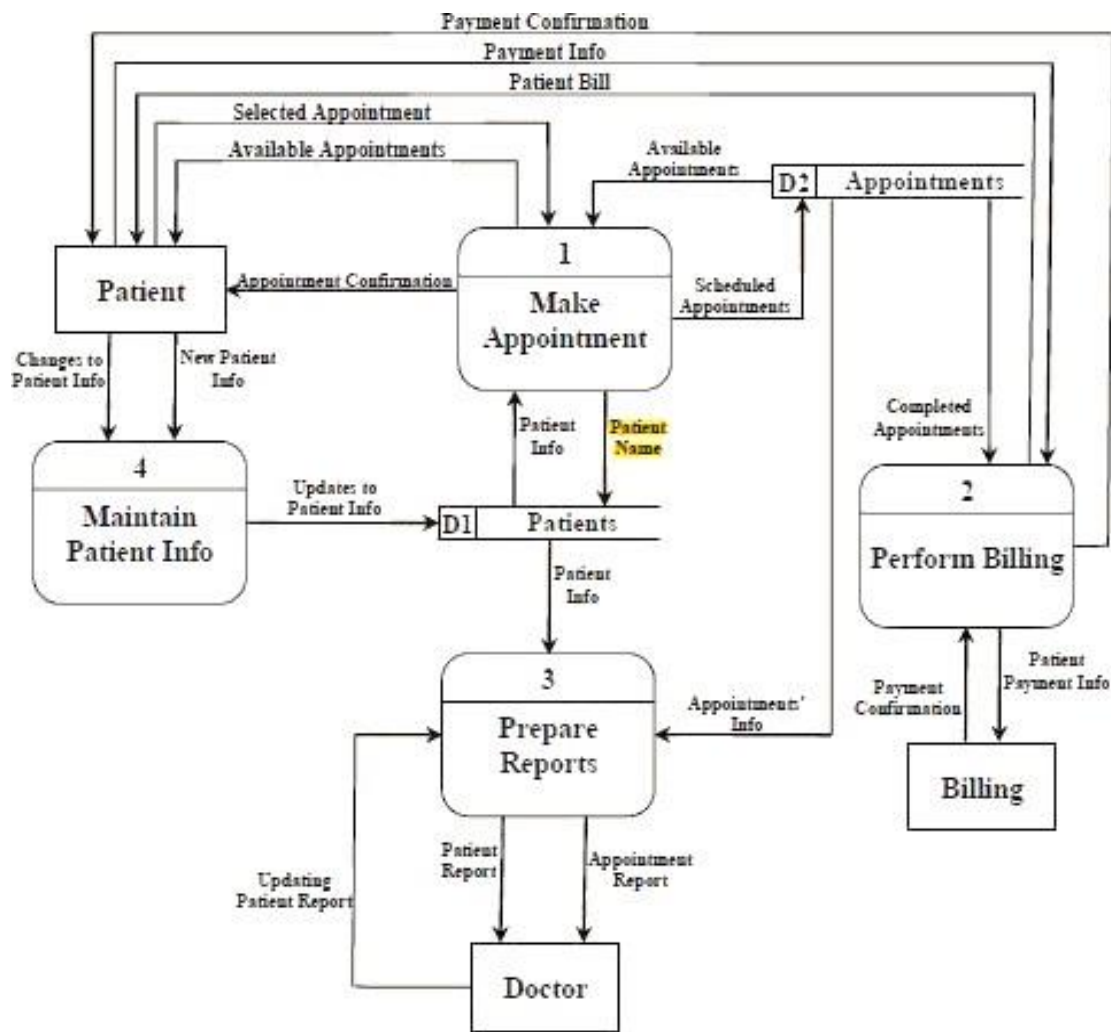


Fig : Level 1 DFD For HMS

2. Object Oriented Approach :

The Object-Oriented Approach (OOA/OOD) is a methodology that models a system as a group of interacting objects—each representing an entity with attributes (data) and methods (behavior). It focuses on real-world modeling and modular development, making systems easier to understand, develop, and maintain.

Here, various type of UML (Unified Modeling Languages) Tools Diagrams are used to display object interactions dynamically to each other as :

- i. **Class Diagrams** : A **Class Diagram** shows the **classes**, their **attributes**, **methods**, and the **relationships** (associations, inheritance, etc.) between them.
- ii. **Object Diagrams** : An **Object Diagram** shows a snapshot of the system at a particular point in time — it represents **instances** of classes and their **relationships**.
- iii. **State Diagrams** : A **State Diagram** shows how an object changes states in response to events. It's useful for modeling lifecycle-driven entities like Patient, Appointment, or Billing.
- iv. **Sequence Diagrams** : A **Sequence Diagram** illustrates **how objects interact in a sequence of messages** over time. It's ideal for modeling specific **scenarios** or **use cases**, like patient registration or appointment booking.
- v. **Activity Diagrams** : An **Activity Diagram** is a type of UML diagram that represents the workflow of activities in a system. It models the sequential flow from one activity to another and is especially useful for visualizing business processes or use case logic.
- vi. **Use case Diagrams** : A **Use Case Diagram** is a type of UML diagram that visually represents the **interactions between users (actors)** and the **system (use cases)**. It helps define the **functional requirements** of the system.

Describing all types of UML Diagrams to represent entities and their relationships in Hospital Management System (HMS) :

Class Diagram : From given figure, it consists of total of 7 classes named Medicine, Transactions, Physician, Patient, Admin, Prescription & Admission. . It defines entities, their attributes, and operations (methods) illustrating how objects like **Patient, Admin, and Prescription** interact in the system.

Where **Admission** has a composition relationship with **Prescription**, indicating a strong lifecycle link (prescriptions exist as part of an admission process) . **Admin** is associated with several classes, meaning Admin can manage or interact with them. **Transactions** are connected with **Patient**, likely tracking payments or services used .

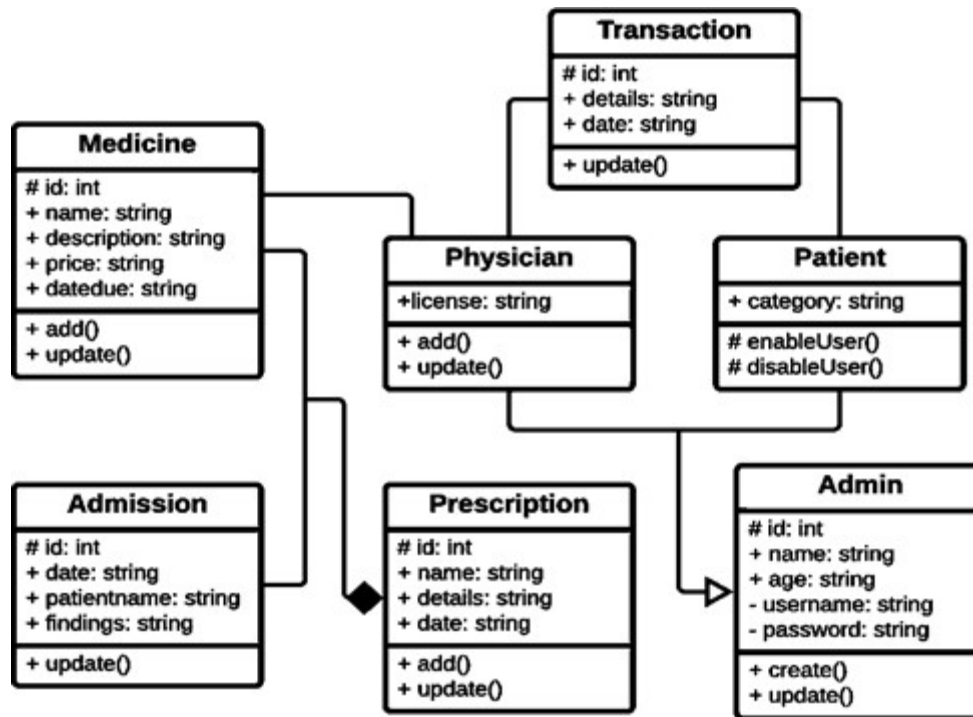


Fig : Class Diagram for HMS

Use Case Diagram : Here, the given diagram represents the actors and their use cases.

1. Administrator

Responsible for system and hospital management.

- Manage Users (Doctors, Nurses, Patients)
- Manage Appointments
- Generate Reports

2. Doctor

Handles medical diagnosis and patient care.

- View Patient Records
- Diagnose Patient
- Prescribe Medicine

3. Nurse

Assists doctors and cares for admitted patients.

- View Assigned Patients
- Update Vitals
- Assist in Tests

4. Patient

Accesses personal and medical information.

- Register & Login
- Book Appointment
- View Prescription
- Make Payment

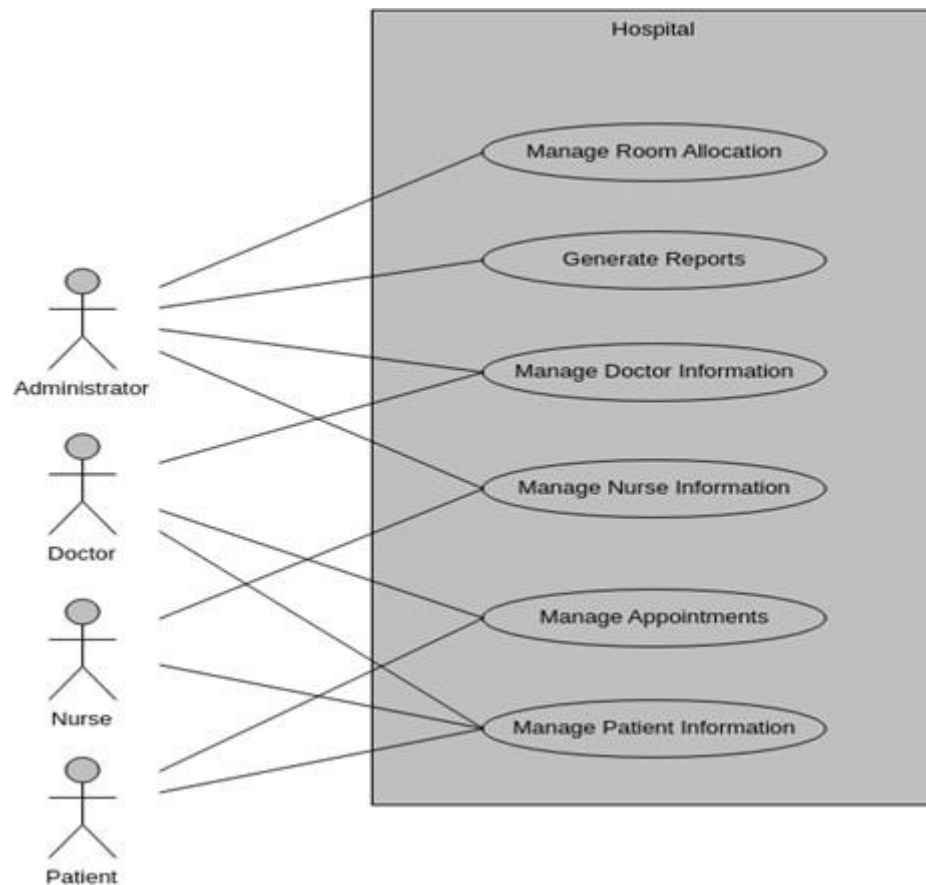


Fig : Use-Case Diagram for HMS

State Diagram : The following diagram models the **state transitions** of a hospital's **patient care and management system**, tracking how a patient progresses through various stages of care and system processing.

The system begins with the **Patient** entering or initiating interaction (e.g., registration, arrival at hospital) & at the end, it finally reached after the **Patient List** state, possibly indicating patient discharge, final record update, or transition to another department.

Hospital Control System State Diagram

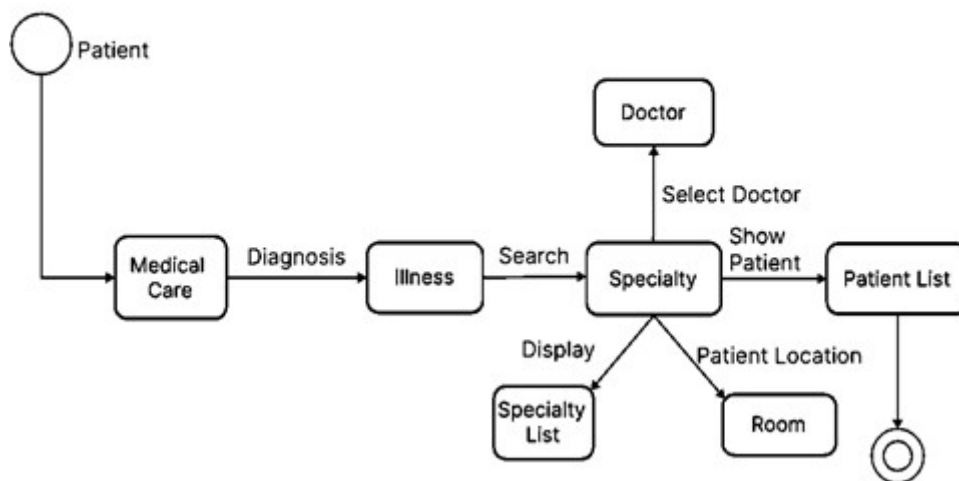


Fig : State Diagram For HMS

Sequence Diagram : The following sequence diagram models the interactions between various entities in a hospital setting: **Patient**, **Receptionist**, **Patient Database**, and **Ward**. It shows the step-by-step communication and data flow when a patient registers and is assigned a ward.

Actors / Objects Involved

1. **Patient** – Initiates the registration process.
2. **Receptionist** – Facilitates data entry and ward assignment.
3. **Patient Database** – Stores and retrieves patient information.
4. **Ward** – Maintains ward availability status and assigns beds.

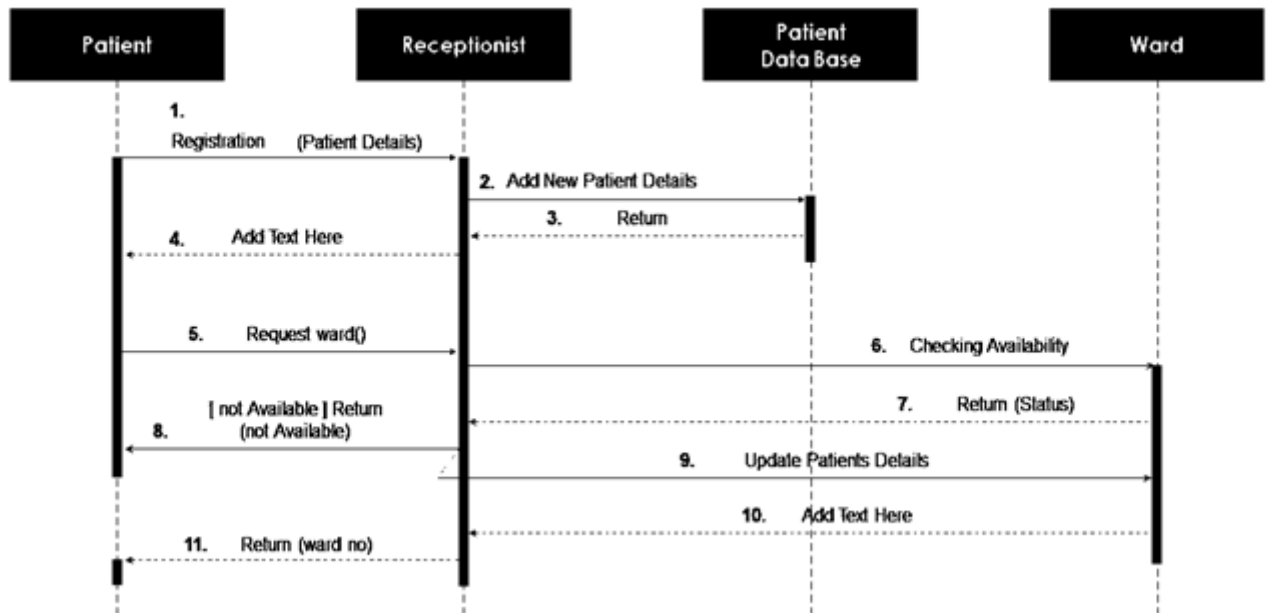


Fig : Sequence Diagram for HMS

Component Diagram : The **Component Diagram** below shows the structural relations between components in an online Hospital Management System. The connected components by dotted lines represent relationships within the systems.

In the diagram, It can be seen that there total of six components name **Patient, Appointment, Receptionist, Doctor, Appointment List, and Patient List.**

Here, **Admin** can add / delete / edit **Doctor details, and Receptionist details.** Admin can view different users list. **Doctors** can **view the appointment in the appointment** and patients' list. A **receptionist** can add / view the **details of patients** on the patient list. Admin can add / delete / edit appointments in the appointments list.

Fig : Component Diagram for HMS

Chapter 3

SYSTEM DESIGN

3.1 Database Design

3.2 Forms and Report Design

3.3 Interface and Dialogue Design

3.1 System Design

System Design is the process of defining the architecture, components, modules, interfaces, and data flow for a system to satisfy specified requirements. In the context of a Hospital Management System, it focuses on designing how different hospital operations interact via software.

Major Design Considerations in Hospital Management are :

- **Scalability:** Should support expansion for more departments.
- **Security:** Must protect sensitive health data (e.g., encryption, access control).
- **Reliability:** 24/7 uptime for critical hospital operations.
- **User-Friendly UI:** Easy for medical and non-technical staff to use.
- **Backup & Recovery:** In case of data loss or power failure.

Basically, there are three important design considerations are made under Hospital Management System: Database Design, Forms and Report Design and Interface and Dialogue Design.

1. Database Design :

Database design is the process of structuring a database schema that stores and manages data efficiently and ensures data integrity and security.

While Performing Designing on Database, the important crucial task we have to is we have to transform the **ER Diagram** of above Hospital Management System (HMS) into the **Relational Table** (also called schema design). Each **entity and relationship** is converted into a table with **Primary Keys (PK)** and **Foreign Keys (FK)** where applicable.

1. For Patient :

No.	Column Name	Data Type	Description
1.	P_ID(PK)	INT/VARCHAR(50)	Patient ID
2.	Name	VARCHAR(50)	Full Name
3.	DOB	DATE	Date of Birth
4.	Age	INT	Age of Patient
5.	Gender	VARCHAR(30)	Male/Female/Other
6.	Mob_No	INT	Mobile Number

2. For Doctor :

S.No.	Column Name	Data Type	Description
1.	E_ID(PK, FK)	INT	Employee ID(From Employee)
2.	Dept	VARCHAR(30)	Department
3.	Qualification	VARCHAR(30)	Doctor's Degree or Qualification

3. For Nurse :

S.No.	Column Name	Data Type	Description
1.	E_ID(PK, FK)	INT	Employee ID (from Employee)

4. For Receptionist :

S.No.	Column Name	Data Type	Description
1.	E_ID(PK, FK)	INT	Employees ID (from Employee)

5. For Employee :

S.No.	Column Name	Data Type	Description
1.	E_ID(PK)	INT	Unique Employee ID
2.	Name	VARCHAR(30)	Name of Employee
3.	Salary	DECIMAL	Monthly Salary
4.	Sex	VARCHAR(30)	Gender (M/F)
5.	Mob_No	INT	Phone Number
6.	Address	VARCHAR(35)	Full Address
7.	City	VARCHAR(35)	City
8.	State	VARCHAR(35)	State
9.	Pin_No	INT	Postal Code (ZIP)

6. For Bills :

S.No.	Column Name	Data Type	Description
1.	B_ID(PK)	INT	Bill ID
2.	Amount	DECIMAL	Amount Billed
3.	P_ID(FK)	INT	Patient ID (From Patient)

7. For Test_Report :

S.No.	Column Name	Data Type	Description
1.	R_ID(PK)	INT	Report ID
2.	Test_Type	VARCHAR(30)	Type of Medical test
3.	Result	VARCHAR(30)	Test Result
4.	P_ID(FK)	INT	Patient ID (from Patient)

8. For Rooms :

S.No.	Column Name	Data Type	Description
1.	R_ID(PK)	INT	Room ID
2.	Type	VARCHAR(30)	Type of Room (eg. ICU,General)
3.	Capacity	INT	No.of Beds
4.	Availability	VARCHAR(30)	Status (Available, Occupied)

9. For Records :

S.No.	Column Name	Data Type	Description
1.	Record_No(PK)	INT	Record Number
2.	App_No	VARCHAR(30)	Appointment Number

10. For Consults (Relationship: Patient ↔ Doctor)

S.No.	Column Name	Data Type	Description
1.	P_ID(FK)	INT	Patient ID
2.	E_ID(FK)	INT	Doctor's Employee ID

11. For Assigned (Patient ↔ Room)

S.No.	Column Name	Data Type	Description
1.	P_ID(FK)	INT	Patient ID
2.	R_ID(FK)	INT	Room ID

12. For Governs (Nurse ↔ Room)

S.No.	Column Name	Data Type	Description
1.	E_ID(FK)	INT	Nurse's Employee ID
2.	R_ID(FK)	INT	Room ID

13. For Maintains (Receptionist ↔ Records)

S.No.	Column Name	Data Type	Description
1.	E_ID(FK)	INT	Receptionist Employee ID
2.	Record_No(FK)	INT	Record No

2. Forms and Report Design :

Forms are the interfaces used for data entry, editing, and interaction. They should be simple, consistent, and role-based (Admin, Doctor, Nurse, Receptionist).

Below are describing common forms for different entities involved in Hospital Management System (HMS) :

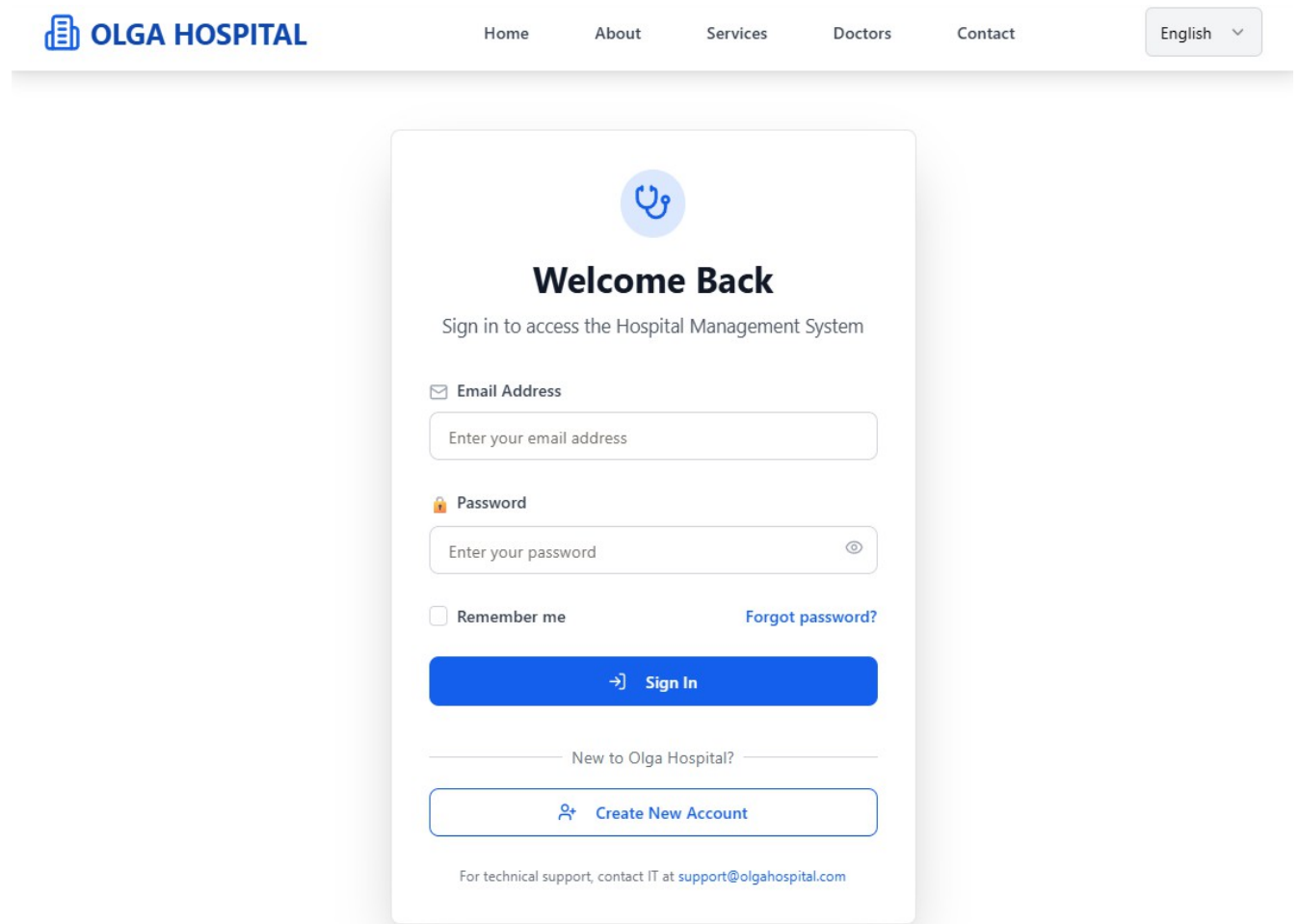
Form Name	Description	Fields / Controls
Patient Registration	Adds a new patient to the system.	Name, DOB, Gender, Phone, Address, Medical History
Doctor Profile	Adds or edits doctor information.	Name, Specialization, Phone, Email, Department
Appointment Form	Schedules patient-doctor appointments.	Patient ID, Doctor ID, Date, Time, Status
Billing Form	Generates invoice based on services.	Patient ID, Services Rendered, Charges, Payment Method
Login Form	Authenticates users based on role.	Username, Password
Vitals Entry (Nurse)	Inputs patient vitals during checkups.	Patient ID, Temperature, Blood Pressure, Notes

Similarly, **Reports** provide data summaries and insights for decision-making, auditing, and monitoring. **The given table diagram depicts the different reports managed under HMS :**

Report Title	Description	Filters / Parameters
Daily Appointments Report	Lists all scheduled appointments for a given day.	Date
Patient History Report	Shows all records related to a particular patient.	Patient ID or Name
Doctor Activity Report	Summarizes consultations, appointments, and notes by doctor.	Doctor ID, Date Range
Billing Summary Report	Total bills generated within a period.	Date Range, Patient, Payment Type
Room Occupancy Report	Shows current usage status of all hospital rooms.	Floor, Type (ICU, General, etc.)
Medicine Inventory Report	Stock details and expiry status of medicines.	Category, Expiry < Date

Some Samples Pictures :

1. Using Figma Designing Tool, The given image depicts Login Interface for admin using Email and Password for OLGA Hospital Under Hospital Management System.



The image shows a web browser displaying the OLGA HOSPITAL website. The header includes the hospital logo, navigation links (Home, About, Services, Doctors, Contact), and a language selector set to English. The main content area features a login form titled "Welcome Back" with a stethoscope icon. The form includes fields for Email Address and Password, a "Remember me" checkbox, a "Forgot password?" link, a "Sign In" button, and a "Create New Account" button. A footer note provides technical support contact information.

OLGA HOSPITAL

Home About Services Doctors Contact English

Welcome Back

Sign in to access the Hospital Management System

Email Address

Enter your email address

Password

Enter your password

☐ Remember me [Forgot password?](#)

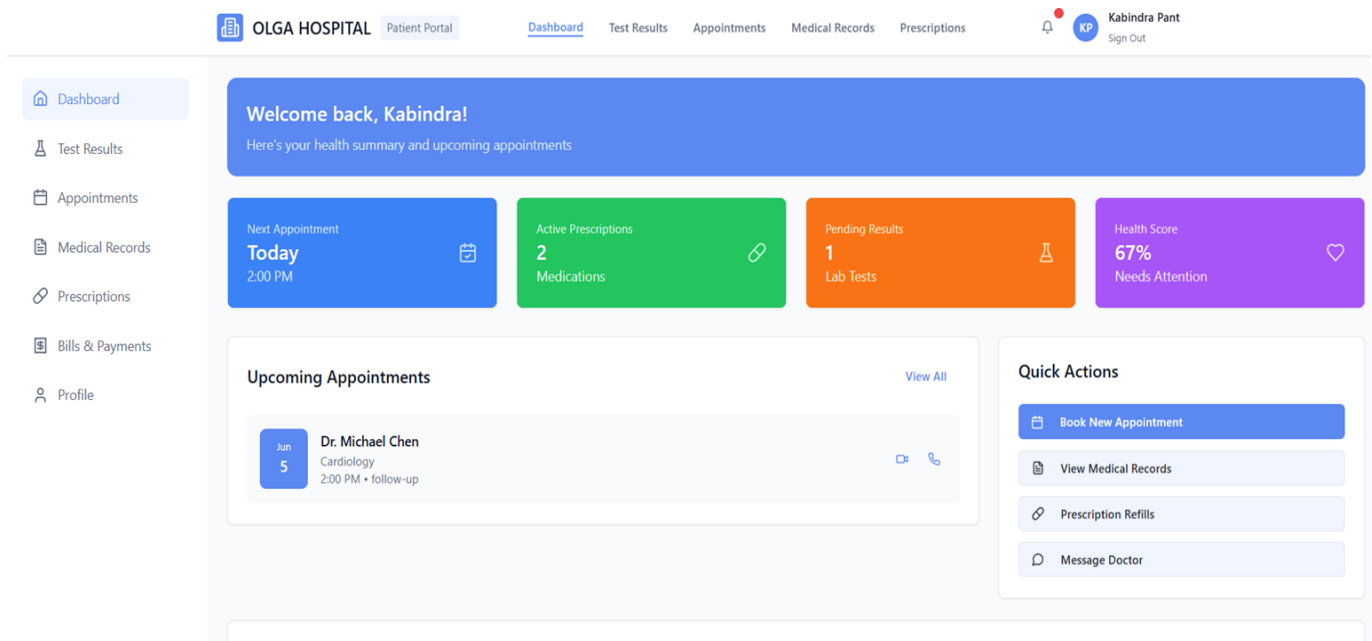
[Sign In](#)

New to Olga Hospital?

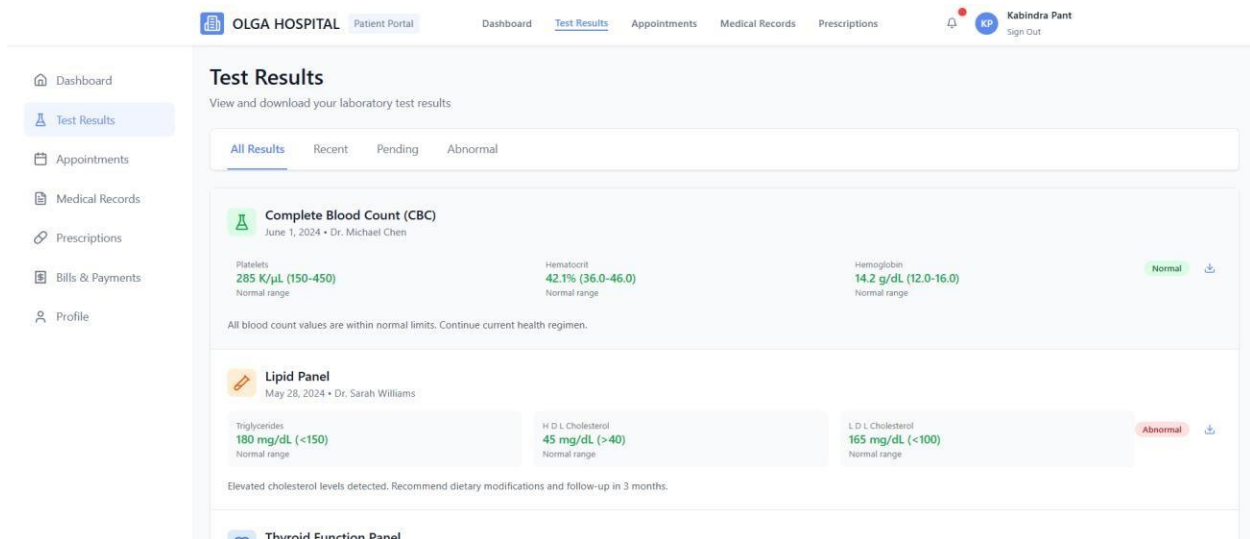
[Create New Account](#)

For technical support, contact IT at support@olgahospital.com

2. Patient Dashboard Within the OLGA Hospital Mangement System, personalized for a user named **“Khabindra”** . This provides a summary of the patient’s health status and upcoming appointments.



3. The given two images displays a **laboratory test results dashboard** within the OLGA HOSPITAL management system and the other one conveys the profile information of username “Kabindra”. It allows patients to view, interpret, and manage their medical test reports, with sections for result summaries, status tracking, and health metric panels.



OLGA HOSPITAL

Patient Portal

Dashboard

Test Results

Appointments

Medical Records

Prescriptions

KP

Kabindra Pant

Sign Out

Dashboard

Test Results

Appointments

Medical Records

Prescriptions

Bills & Payments

Profile

Profile

Manage your personal information and preferences

Personal Information

First Name

Kabindra

Last Name

Pant

Date of Birth

Not provided

Gender

Not provided

Phone Number

Not provided

Email Address

harmocsit@gmail.com

Address Information

Street Address

Not provided

Profile Photo

KP

Change Photo

Health Information

Blood Type

Not provided

Height

Not provided

Weight

Not provided

4. The given image presents the administrator dashboard for *OLGA HOSPITAL Management System*, providing hospital administrators with a comprehensive overview of operations, patient data, and quick access to key functions. The Dashboard describes a merit panel including total patients **test point of 2847** with monthly growth of 12% and total today's appointments of **43 where 8 are completed**.

OLGA HOSPITAL

Admin Portal

Dashboard

Patients

Appointments

Billing

Medical Records

Doctors

Settings

Dashboard Overview

Welcome back, Admin Johnson

Search patients, appointments...

3

Total Patients

2,847

+12% from last month

Today's Appointments

43

8 completed

Pending Bills

\$89,247

15 overdue

Active Staff

127

98% attendance

Recent Patients

View All

PATIENT	PHONE	LAST VISIT	STATUS
SJ Sarah Johnson ID: P-2024-0156	+1 (555) 123-4567	2024-01-15	Active
MC Michael Chen ID: P-2024-0157	+1 (555) 987-6543	2024-01-12	Pending
ER Emily Rodriguez ID: P-2024-0158	+1 (555) 456-7890	2024-01-10	Active

Quick Actions

Add New Patient

Schedule Appointment

Generate Bill

View Reports

5. This section displays a **patient registration form** for OLGA HOSPITAL Management System, featuring both English and Greek language fields for capturing **new patient information**.

OLGA HOSPITAL
Admin Portal

Dashboard Overview
Welcome back, Admin Johnson

Patients
Manage patient records

Search patients by...

Add New Patient

First Name:

Last Name:

Date of Birth:

Gender:

Phone Number:

Email:

Emergency Contact:

Emergency Phone:

Address:

Cancel Add Patient

6. The following dashboard depicts the number of doctors including different fields of Cardiology, Dermatology, Orthopedics, Neurology to test different type of patients.

OLGA HOSPITAL
Admin Portal

Dashboard Overview
Welcome back, Admin Johnson

Doctors
Manage medical staff and their information

Search doctors by name, specialization...

Add New Doctor

Doctor	Department	Contact	Status	Actions
Dr. Michael Chen Cardiology	Cardiology Department	m.chen@olgahospital.com +1 (555) 123-4567	active	Edit Delete
Dr. Emily Rodriguez Dermatology	Medical Ward	e.rodriguez@olgahospital.com +1 (555) 234-5678	active	Edit Delete
Dr. David Thompson Emergency Medicine	Emergency Department	d.thompson@olgahospital.com +1 (555) 345-6789	active	Edit Delete
Dr. Sarah Williams Pediatrics	Pediatric Ward	s.williams@olgahospital.com +1 (555) 456-7890	active	Edit Delete
Dr. James Anderson Orthopedics	Orthopedic Department	j.anderson@olgahospital.com +1 (555) 567-8901	active	Edit Delete
Dr. Lisa Martinez Neurology	Neurology Department	l.martinez@olgahospital.com +1 (555) 678-9012	active	Edit Delete

Chapter 4

IMPLEMENTATION **AND TESTING**

4.1 System Testing

4.2 Types of System Testing Enlisting Test Cases

4.1 System Implementation :

Implementing a Hospital Management System (HMS) involves converting the design into a working software system that efficiently manages hospital operations like patient records, billing, room allocation, and staff management.

To Implement different layers of HMS , different Programming Tools and techniques are used as :

2. **Frontend** : HTML, CSS, Javascript, React / Angular
3. **Backend** : Java / Python (Django, Flask) , PHP, Node JS
4. **Database** : MySQL, Oracle
5. **Frameworks** : Laravel (PHP), Spring Boot (Java)
6. **Deployment** : Apache, Docker, AWS

Below representing a simple Database (SQL) and Programming Language (we used PHP) to implement the basic Project Structure of Hospital Management System as :

1. Database Connection (config/db.php)

```
<?php
$host = 'localhost';
$db  = 'hospital_db';
$user = 'root';
$pass = "";
$conn = new mysqli($host, $user, $pass, $db);

if ($conn -> connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```


2. Adding *Patient* (patients/add.php)

```
<?php include '../config/db.php'; ?>

<?php

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $name = $_POST['name'];
    $dob = $_POST['dob'];
    $gender = $_POST['gender'];
    $phone = $_POST['phone'];

    $sql = "INSERT INTO patients (name, dob, gender, phone) VALUES ('$name', '$dob', '$gender', '$phone')";

    if ($conn->query($sql) === TRUE) {
        echo "Patient added successfully";
    } else {
        echo "Error: " . $conn->error;
    }
}

?>

<form method="POST" action="">

    Name: <input type="text" name="name"><br>
    DOB: <input type="date" name="dob"><br>
    Gender: <input type="text" name="gender"><br>
    Phone: <input type="text" name="phone"><br>
    <input type="submit" value="Add Patient">

</form>
```

3. Adding inputs in *patients* Table (Using MySQL)

```
CREATE TABLE patients (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    dob DATE,  
    gender VARCHAR(10),  
    phone VARCHAR(15)  
);
```

4. Login Example (login.php)

```
<?php  
session_start();  
include 'config/db.php';  
if ($_SERVER["REQUEST_METHOD"] == "POST") {  
    $username = $_POST['username'];  
    $password = $_POST['password'];  
    $res = $conn->query("SELECT * FROM users WHERE username='$username' AND  
password='$password'");  
    if ($res->num_rows > 0) {  
        $_SESSION['username'] = $username;  
        header("Location: index.php");  
    } else {  
        echo "Invalid credentials.";  
    }  
}  
?>
```

```

<form method="POST">

    Username: <input name="username"><br>

    Password: <input name="password" type="password"><br>

    <input type="submit" value="Login">

</form>

```

4.2 System Testing :

System testing is a crucial step in the Software Development Life Cycle (SDLC) to ensure that the entire system works as intended. In the case of a Hospital Management System, testing verifies that all features—from patient registration to billing—function correctly, securely, and efficiently.

Here's a Simple Testing Result Sample of Hospital Management System including Test ID, Expected and Actual Result and whether patients are passed or not.

Test Case ID	Description	Expected Result	Actual Result	Status
TC001	Add patient	Patient saved	Patient saved	Pass
TC002	Invalid login	Error message shown	Error message shown	Pass
TC003	Generate bill	Total amount correct	Total correct	Pass
TC004	Book past appointment	Show validation error	Error shown	Pass
TC005	SQL Injection in login	Blocked login attempt	Blocked	Pass

System Testing is conducted to validate the **end-to-end behavior** of the complete and fully integrated software product.

- i) Verify Complete System Functionality.
- ii) Validate Businesses Requirements.
- iii) Ensure Data Flow Accuracy and Test Reliability & Performances.

4.3 Types Of System Testing :

Here are the main types of system testing relevant to your Hospital Management System (HMS) project in PHP:

- **Unit Testing** : Unit testing is a software testing method where individual components or modules of an application are tested in isolation to verify their correctness.

In HMS, It tests individual PHP functions, classes, or modules (e.g., addPatient(), scheduleAppointment()). Used Tools: PHP Unit (for advanced PHP projects). For eg: ***Test if insertPatient() correctly stores patient data in the patients table.***

2. Integration Testing : Integration Testing in a Hospital Management System (HMS) verifies that different modules or components work together correctly, ensuring seamless data flow and functionality across the system. It also ensures that modules work together, such as:

- i. Patient registration → Room assignment
 - ii. Appointment booking → Doctor schedule
- For eg : ***Test if an appointment booked for a patient automatically updates the doctor's availability.***

3. Functional Testing : Functional testing in a Hospital Management System (HMS) involves verifying that each feature of the system operates as intended based on the defined requirements. It ensures that the system can perform its core functionalities, such as patient registration, record management, appointment scheduling, billing, and report generation, accurately and reliably.

Consider a Test Case as :

Functionality	Test Case	Expected Output
Add Patient	Submit valid form	Patient added successfully
Appointment Booking	Select date/time, doctor, patient	Appointment created
Billing Module	Calculate service charges	Total bill generated
Login	Enter correct credentials	Redirect to dashboard
Room Availability	Assign patient when room is free	Room assigned successfully

4. **System Testing** : System testing in a Hospital Management System (HMS) using AI focuses on verifying the overall functionality, performance, and reliability of the integrated AI components within the system.

It is used in testing the **entire HMS application** as a whole. It Simulates real-world hospital workflow to check:

- Performance
- Data flow across modules
- Session management

For eg : **Register** → **Book Appointment** → **Doctor Diagnoses** → **Add Tests** → **Generate Bill** → **Logout**

5. **Acceptance testing** : It is done with hospital staff or supervisors to validate that the HMS meets all **functional requirements**. And also to it can be accepted for deployment.

Different Checklist:

- All modules accessible?
- All user roles functional?
- Reporting and logs working?
- Data backups available?

Here's a Simple Test Cases for Registering a new Patient and Schedule an Appointment :

Field	Value
Test Case ID	UAT-HMS-001
Objective	Ensure that a new patient can be registered and scheduled for an appointment by a receptionist
Preconditions	User (Receptionist) is logged in
Test Steps	1. Go to "Add Patient" screen. 2. Fill out patient details. 3. Submit the form. 4. Navigate to "Schedule Appointment". 5. Select the new patient and assign to a doctor/date/time. 6. Submit appointment
Expected Result	Patient is added to the database- Appointment is successfully scheduled and appears in both patient and doctor dashboards
Actual Result	(To be filled by tester)
Status	Pass / Fail
Tested By	Shishir Pandey
Date	2025-06-04

Chapter 5

CONCLUSION

Working on the **Hospital Management System (HMS)** project has been an enriching and insightful experience for our entire team. Throughout the project lifecycle - from initial planning to final implementation, we gained a deeper understanding of how real-world hospital operations can be translated into efficient digital workflows.

We encountered various challenges, including defining clear use cases, designing an optimized database, managing user access control, and ensuring smooth integration between modules. Solving these challenges helped us strengthen our **problem-solving, programming, and collaboration** skills.

Our team also learned the importance of :

- **Systematic software design**, using tools like ER diagrams, DFDs, and use case models.
- Applying **real-life testing strategies**, such as unit testing and user acceptance testing, to validate functionality.
- Understanding user experience, especially from the perspectives of different roles like doctors, nurses, and admin staff.

By using **PHP and MySQL**, we became more confident in full-stack web development and database design, while also applying practical knowledge of **system analysis and design principles** covered in our coursework.

Most importantly, working as a team helped us improve our communication, task delegation, and time management, skills that are essential for real-world software development projects.

This project helps reduce human error, saves time in administrative work, and enables better data management and security. It also improves communication between hospital departments and enhances patient service quality.